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# Caucus: an authentication protocol for cloud federation

Anant V. Nimkar and Soumya K. Ghosh

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ABOUT

## Abstract

Cloud federation provides computing services of internal and external cloud providers. These computing services (e.g., virtual resources or services etc.) are collectively owned, accessed and controlled by one or more federating participants like user, broker, cloud providers, service provider etc. Thus, subjects are subsets of federating participants for use, execution, deployment and management of computing services. Each such subject must be authenticated before authorisation of computing services. Identity management solutions cannot address authentication of subjects comprising of more than two federating participants. In this paper, we propose a protocol for authentication (called as caucus authentication protocol) of subjects as a subset of one or more federating participants using a variant of multi-party computation (MPC). Theoretical study attempts to prove liveness and safety properties of proposed caucus authentication protocol (CAP) for the validation of dead-lock-free communication. The simulation results show that the protocol provides authentication of subjects in acceptable response time.

## Keywords

mandatory access control, MAC, discretionary access control, DAC, multi-party computation, MPC, IaaS, SaaS, cloud, federation, authentication, security, access control

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## SMART SHOPPING USING BEACONS

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### Abstract

Internet of Things (IoT) is a paradigm shift driven by the recent evolution in the world of sensor networks, mobile equipments, wireless communications and cloud technologies. IoT brings about the possibility to use technology and automation in daily life which opens up possibilities of innovative business models. Bluetooth Low Energy (BLE) is a technology that provides location based services for cost-effective communication within a given range. This paper concentrates on the use of this technology in retail markets. BLE extends the capabilities and accessibility of smartphones and aids retailers to offer services to customers depending on their in-store communication with the products. The proposed system will utilize motion sensing, proximity sensing and machine learning algorithm for personalizing and improving customer experience in retail stores.

**Index Terms:** Internet of Things (IoT), Motion Sensing, Proximity Sensing, Machine Learning, Retail.

### 1. INTRODUCTION

The Internet of Things(IoT) is a standard of communication in which everyday objects are equipped with sensors. IoT is being put into effect in every domain of human concern like traffic navigation, security, smart agriculture, home automation, retail and healthcare. Beacon Technology is one such aspect of IOT. Retail is currently the most popular area in which beacon technology is being implemented. Beacons provide a modern way for interaction between customers and retailers in a store.

Due to the increasing popularity of online shopping, offline retail stores are facing threat of being obsolete. As a result, retailers are making

an attempt to integrate technology in offline shopping. Mobile devices play a significant role in connecting retailers to their customers. Understanding the behaviour of the customer within the retail store is crucial for providing ideal shopping experiences, ensuring optimal layout, and developing an efficient operational model. Marketing to customers at the right time based on their location has the potential to cause a paradigm shift in retailing.

Bluetooth Low Energy is a wireless technology for short range communication requiring low power consumption. BLE devices can be used to monitor and track other devices within range and this capability can be used in retail to improve customer experience. A mobile application design to interact with the beacons offers a number of benefit to the customers as well as retailers. Since beacons are designed for mass distribution and operate on a coin-cell battery, these tiny devices can be placed throughout a retail store at walls or on products enabling customers via the application to be notified of recommendations, greetings, special offers when they enter the store or provide specific product information of items depending on their in-store actions or proximity with the products.



Fig. 1. Working of BLE beacons (Source : [www.empresariozcapital.com/files/6214/3292/6082/beacons1.jpg](http://www.empresariozcapital.com/files/6214/3292/6082/beacons1.jpg))



# Analysis of subthreshold SOI FinFET based two stage OTA for low power

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## Abstract

In this paper, subthreshold design and analysis of Silicon on Insulator Fin Field Effect Transistor (SOI FinFET) based two stage Operational Transconductance Amplifier (OTA) is presented for low power and low supply voltage in nanometre regime. The OTA design optimization is achieved by  $g_m/I_D$  methodology which helps to determine the device aspect ratios. Compactness is achieved by using nanometre FinFET technology. The OTA design is simulated using 30 nm SOI FinFET Berkeley Short-channel IGFET Common Multi-gate (BSIM-CMG) model, with bias current and supply voltage of 20 nA and  $\pm 0.5V$ ; respectively. The simulation results in subthreshold regime of FinFET based two stage OTA has a gain of 57 dB with a phase margin of 69.81 degree, Common Mode Rejection Ratio (CMRR) of 61.55 dB and power consumption of 108 nW.

**Keywords** SOI FinFET · Subthreshold · Low supply voltage · OTA · Gain · CMRR · Low Power dissipation

## 1 Introduction

The wide spread propagation of passive sensors facilitated by the Internet of things revolution and requirement of low-power biological sensors has spurred the interest in the field of subthreshold region operating devices. These devices must support long battery life as they will be deployed in sensors which may be located in sparsely populated, inaccessible regions of the world. An improvement for new technologies on numerous fronts, both on device and circuit level, systems and applications is necessary. Although continuous down scaling of supply voltage and transistor channel-length has unbelievably run-down the performance of the analog part of the circuit. It can be concluded that the nanometre FinFET device

technology for miniaturization is the best option, for the low power regime in analog domain [1–3]. The analog circuits operating in subthreshold region are used for low voltage and low power application because they have high  $g_m/I_D$  ratio, providing high energy efficiency. Moreover, most of the research work reported in analog domain is based on nanometre CMOS technologies [4–10]. This paper deals with current research in analog domain, related to design of subthreshold FinFET based OTA particularly at low technology node of 30 nm.

The FinFET device is biased in the subthreshold region of operation also known as weak inversion region. This ensures that the FinFET draws very low current and also dissipates negligible amount of power. The design of an ultra-low power highly sensitive even at low voltages analog front end is a must in biomedical applications. Hence our focus was on OTA which is an integral part of any analog circuit. It can be used as a part of the front end of biomedical devices as well as used in passive sensors in remote places. In this paper we investigate and simulate the subthreshold region performance of 30 nm SOI FinFET based two stage Operational Transconductance Amplifier (OTA) with a high- $\kappa$  gate dielectric hafnium oxide ( $HfO_2$ ) for low power and low voltage supply applications. The OTA circuit consists of input differential stage followed by

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# Communication

## A Compact Dual Band Gap Electromagnetic Band Gap Structure

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**Abstract**—In this communication, compact dual band gap two via double slot type electromagnetic band gap (TVDS-EBG) structure is presented. Analysis of the TVDS-EBG structure is done by using the LC model. Two resonance parallel LC circuit per unit cell results dual band gap in x-and y-direction. Compactness is achieved because of two via per unit cell, square slot, and inverse C-type slot on EBG patch. The simulation of the unit cell TVDS-EBG is carried out using eigenmode solution of Ansys high frequency structure simulator (HFSS). Simulated results are validated by experimental results. Simulated and experimental results prove that compared to two-layer edge located via inter digital-EBG, edge located via-EBG, and minkowski EBG, TVDS-EBG gives size reduction of 20.48%, 29.03%, and 31.25% respectively with simple structure. Application of proposed dual band gap TVDS-EBG for ultra wide band (UWB) monopole antenna is also demonstrated. Simulated and measured results shows that dual band-notched can be obtained by using a unit cell of TVDS-EBG at fed line of an elliptical monopole antenna.

**Index Terms**—Dispersion diagram, dual-band gap electromagnetic band gap structure, monopole antenna, size reduction, ultra-wide-band communications (UWB).

### I. INTRODUCTION

ELECTROMAGNETIC band gap (EBG) structures are periodic cells realized by arrangement of dielectric substrate, and metallic conductors [1]. These periodic structures are very useful in RF, and microwave community due to their unique characteristics. EBG structures are used to improve the performance of antenna, array, and radar cross section (RCS) reduction [2]-[6]. Recently many compact EBG structures are reported [3]-[9]. EBGs with dual band gap are required in some wireless applications like global positioning system (GPS), wireless local area network (WLAN) (5.15 to 5.35 GHz, 5.725 to 5.825 GHz), and personal communications (GSM900/DCS1800). In recent years, many dual band gap EBG structures are presented. In [10] a tunable PIN diode based EBG structure for dual band gap application is investigated. Shuai Zhang et al [11] presented a two-layer edge located via and interdigital EBG (TELI-EBG) for RCS reduction with unit cell size  $0.083\lambda_{5.0GHz}$ . However, the realization of these EBG is quite difficult. X. L. Bao et al [12] presented a dual band gap fractal type EBG for GPS patch antenna application with unit cell size  $0.112\lambda_{1.24GHz}$ . T. Masri et al [13] presented a dual-band microstrip array using conventional mushroom type EBG (CMT-EBG) and Minkowski type EBG with unit cell size  $0.096\lambda_{2.4GHz}$ . Long Li et al [14] presented dual band gap simple EBG structure with unit cell size of EBG  $0.152\lambda_{9.15GHz}$ . However these

presented dual band gap EBG designs have failed to have simple structure [10] [11], and compact size [12]- [14]. In this work, we have presented design, and analysis of a compact, dual band gap, and single layer two via double slot type electromagnetic band gap (TVDS-EBG) structure with its application. A compactness, and dual band gap are achieved because of two via per unit cell, square strip, and slot on inner patch. Section II presents a proposed dual band gap EBG prototype, its design, and dispersion diagram. In section III, fabricated TVDS-EBG cells, band gap measurements using suspended microstrip line method (SML) has been explained. Application of TVDS-EBG structure is demonstrated in section IV. Last section is concluded by comparing results for TVDS-EBG with reported dual band gap EBG structures.

### II. PROTOTYPE OF TVDS-EBG, AND DISPERSION DIAGRAM

EBG structures are modeled as a parallel LC equivalent circuit when the patch size is small compared to operating wavelength ( $\lambda_0$ ) [1]. Surface impedance ( $Z_s$ ) and resonance frequency ( $f_c$ ) of the EBG structure are given as [1] [2]

$$Z_s = \frac{j\omega L}{1 - \omega^2 LC} \quad (1)$$

and

$$f_c = \frac{1}{2\pi\sqrt{LC}} \quad (2)$$

The inductance (L) and Capacitance (C) are calculated as [1] [2]

$$L = \mu_0 h \quad (3)$$

and

$$C = \frac{w\epsilon_o(\epsilon_r + 1)}{\pi} \cosh^{-1}\left\{\frac{2w + g}{g}\right\} \quad (4)$$

with band gap bandwidth [1]

$$BW = \frac{\Delta\omega}{\omega} \quad (5)$$

where, ( $\mu_0$ ) = permeability of free space, and (h) = substrate height, (w) = width of each EBG patch, (g) = gap between two EBG cell, ( $\epsilon_r$ ) = dielectric constant of the substrate, and ( $\epsilon_0$ ) = permittivity of free space. In order to design a dual band gap EBG structure, two series resonant circuit per unit cell are required. The prototype of proposed TVDS-EBG, which is an extension work of two via slot type-EBG (TVS-EBG) [9] with simple inverse C-type slot on inner square patch as shown in Fig. 1. TVDS-EBG cell is equivalent to two series resonant circuits as shown in Fig. 2 (a) and (b) results in dual band gap. As shown in Fig. 2 (a),  $C_1$ , and  $C_3$  are formed due to fringing effect between outer slot, an inner square patch.  $C_2$ , and  $C_4$  are due to inverse C-type slot on an inner patch. Inductance

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# A Compact Two-via Hammer Spanner-Type Polarization-Dependent Electromagnetic Bandgap Structure

Pramod P. Bhavarthe<sup>1</sup>, Surendra S. Rathod, *Member, IEEE*, and Kuraparthi T. V. Reddy, *Senior Member, IEEE*

**Abstract**—In this letter, a compact two-via hammer spanner-type polarization-dependent electromagnetic bandgap (TVHS-PDEBG) structure is proposed. Asymmetrical structure of the TVHS-PDEBG results in different bandgap properties in the  $x$ - and  $y$ -directions. Compactness of the structure is achieved because of the usage of two-via per unit PDEBG cell, U-shaped rectangular slot, stretched strip, and square EBG patch with central located via embedded in adjacent PDEBG cell. The simulation of the TVHS-PDEBG is carried out using Ansoft high-frequency structure simulator, and it is validated experimentally. These results are compared with fabricated central located via EBG having similar parameters. Simulated and experimental results prove that, compared with complementary split ring resonator-based PDEBG and dual asymmetrical U-shaped-type PDEBG, the TVHS-PDEBG presents a 35.48% and 16.66% reduction in the center frequency of the lower bandgap.

**Index Terms**—Dispersion diagram, polarization-dependent electromagnetic bandgap (PDEBG) structure, size reduction.

## I. INTRODUCTION

ELECTROMAGNETIC bandgap (EBG) structures are very popular in RF and microwave community due to their size and applications derived from stopband characteristics. In recent years, several EBG structures have been reported [1]–[3]. It has been investigated that period of EBG should be half of the wavelength at stopband center frequency, hence the EBG structures have large size at the lower frequency [3]. To solve this problem, many compact square EBGs were reported [4]–[6]. In many RF applications, polarization-dependent EBG (PDEBG) structures are required. Yang and Rahmat-Samii [7] reported slot-loaded, rectangular, and offset via PDEBG surfaces with a unit cell size of  $0.16\lambda_{3.0}$  GHz and utilized it to enable a single dipole antenna to radiate circular polarization pattern with a low-profile configuration. Peng *et al.* [8] presented a complementary split ring resonator (CSRR)-based PDEBG with a unit cell size of  $0.093\lambda_{3.82}$  GHz for multibandgap applications. Ullah *et al.* [9] presented the PDEBG structure with a slanted sheet via for linear-to-circular polarization. The unit cell size of the PDEBG for  $0^\circ$  inclination angle is  $0.169\lambda_{2.2}$  GHz. Yi and Qu [10] reported a dual-bandgap, and rectangular PDEBG unit cell with a size of  $0.060\lambda_{3.3}$  GHz  $\times$   $0.077\lambda_{3.3}$  GHz for dual-band left-handed circular polarization (LHCP) application. Han *et al.* [11] introduced

the analysis and design of a slot-type PDEBG to enhance the gain and reduce the radar cross section of patch antenna with a unit cell size of PDEBG  $0.20\lambda_{3.26}$  GHz. These reported PDEBGs give moderate unit cell size reduction. In this letter, we presented the design and analysis of two-via hammer spanner-type polarization-dependent electromagnetic bandgap (TVHS-PDEBG) structure which is compact than the reported PDEBG structures and having frequency variation capability. Compactness of the structure is achieved because of the usage of two via per unit PDEBG, U-shaped slot, and stretched strip embedded in adjacent PDEBG. Section II presents a proposed PDEBG prototype, its design, and reflection phase diagram. In Section III, fabricated TVHS-PDEBG cells and bandgap measurements have been explained. Application of TVHS-PDEBG as a polarization reflector is demonstrated in Section IV. Section V concludes by comparing results for TVHS-PDEBG with the reported PDEBG structures.

## II. PROTOTYPE OF PDEBG AND REFLECTION PHASE DIAGRAM

The surface impedance  $Z$  of the EBG surface is given [1]

$$Z = \frac{j\omega L}{1 - \omega^2 LC} \quad (1)$$

The EBG surface is inductive at low frequency and supports TM waves. At high frequency, it is capacitive and supports TE waves.  $Z$  is very high at resonance frequency ( $\omega_0$ ) [1]

$$\omega_0 = \frac{1}{\sqrt{LC}}. \quad (2)$$

The inductance ( $L$ ) is due to current flowing through via, ground plane, and adjacent EBG unit. The capacitance ( $C$ ) is due to the gap effect between the adjacent EBG unit. From (2), increase in  $L$  or  $C$  of the  $LC$  equivalent circuit results compactness in EBG unit cell. Central located via EBG (CLV-EBG) [1], [2] structure is with wider bandgap and moderate size. To get PDEBG, an asymmetrical EBG structure is required. The prototype of the proposed TVHS-PDEBG is shown in Fig. 1, which is an extension of an forklike EBG [4] and two-via slot-type EBG [6]. Each unit cell of this EBG consists of a square patch with a U-shaped rectangular slot and a stretched strip, including a rectangular patch at its end, embedded in an adjacent EBG. This rectangular patch is formed with a square slot and contains another square patch with a via located at the center. There is another via present at the middle of the stretched strip. Both vias are connected to ground plane. To get compactness, two-via concept is used, which results in increase of the equivalent value of  $L$  per unit EBG cell. Additional equivalent  $C$  is caused due to the edges

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## A Hybrid Approach for MRI Based Statistical Feature Extraction to Detect Brain Tumor

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**Abstract**—Undesired abnormal growth of cells in the brain is a serious neurological problem which is known as brain tumor. Early detection and diagnosis of tumor from an MRI image is a primary task. It is tedious and time consuming task which requires expertise of radiologists. To overcome this limitation, an automated state-of-the-art system is required which can give an estimate about whether the brain MRI image has tumor or not. The literature reveals that feature extraction is a significant step for detection of tumor. In this study, feature extraction is incorporated with significant feature identification so as to minimize the time required for processing the data and to improve the accuracy. This paper presents a technique for statistical feature extraction by hybridization of analytical and algorithmic means and Gray Level Cooccurrence Matrix (GLCM) properties. This gives a contribution in feature extraction which is fundamental for any decision algorithm to give better and accurate results. The experimental results have been evaluated for axial, coronal and sagittal views of brain MRI image. The results show that the proposed method is effective for extracting the significant features.

**Index Terms**—Brain tumor, Magnetic resonance imaging (MRI), feature extraction

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### I. Introduction

Brain tumor or intracranial neoplasm is a mass of involuntary growth of cells which grow in the brain [1]. There are two types of tumor namely benign and malignant tumor. Benign tumor is non-cancerous as it has slow growth and does not spread whereas malignant tumors are cancerous as it can spread fast to other parts of brain and central nervous system [2]. Thus malignant tumors are much more harmful. Meningioma, papilloma, lipoma, adenoma are classified under benign tumors, whereas, sarcoma, carcinoma, lungs tumor comes under malignant tumor. The lifespan of patients with benign tumor is 10 to 12 years. Patients with malignant tumor can survive not more than 14 months even if the treatment is going on. Early detection and diagnosis of brain tumor is not only important for treatment and curing but also for assessment of percentage growth of tumor and follow-ups. Surgical operations, chemotherapy, radiation therapy and so on can be used for the treatment. MRI scan, CT scan, PET scan and various other scans can be used to detect the tumor. MRI is having an additive advantage because it does not emit ionizing radiation rays. As such, no biological hazards have been reported with the use of MRI [3].

Detection of tumor from a brain MRI image is thus a very important step but is complicated because of the complexities in the MRI image arising due to the shape, size, location and distribution of tumor cells. Thus an automatic system will assist the radiologists for easy detection of tumor as well as reduce the complexities. Statistical features like mean, variance, standard deviation, correlation, contrast, energy, entropy, homogeneity etc can be used for feature extraction of brain MRI image and thus can be used to distinguish between a normal brain MRI image and a tumorous image.

Rest of the paper has been organized as follows. Section II describes the Brain tumor and MRI overview which includes types of MRI scans and brain tumor detection factors. Section III describes the process of brain tumor detection using statistical features. Section IV describes Results and Discussion, followed by Conclusion in Section V.

### II. Brain Tumor and MRI Overview

## Computation and Analysis of Excitatory Synapse and Integrate & Fire Neuron: 180nm MOSFET and CNFET Technology

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**Abstract :** Neuromorphic circuits and systems are mixed analog and digital hardware implementations of biological networks. The Neurons and Synapses form the basic building blocks of these networks analogous to their biological counterparts. The analog VLSI implementation of neuron and synapse circuits with devices working in subthreshold regime ensure compactness and low power consumption in these circuits. The use of advanced technologies of MOSFETs has further improved the area and power efficiencies of the neuromorphic circuits. In this work, we have implemented a Static Excitatory Synapse circuit and an Integrate and Fire Neuron circuit in 180nm technology. The circuits are simulated using Cadence software and verified for the functionality with varying circuit parameters. Further, the Static Excitatory Synapse circuit is cascaded to an Integrate and Fire Neuron circuit and simulated with adjusted component and parameter values in 180nm MOSFET technology in Cadence software and the Carbon Nanotube Field Effect Technology in HSPICE software as well. The effect of the parameter values on the functionality of the cascaded circuit and the passage of signals is studied, analyzed and compared in the two technologies in terms of power and area efficiencies of the circuits.

**Keywords**—Carbon Nanotube, Neuromorphic, Neuron, Synapse.

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### I. INTRODUCTION

Neuromorphic Engineering is a branch of Electronics Engineering that deals with the design and architecture of electronic circuits and systems inspired from the biological networks. Carver Mead initiated the investigation in the field of building up of circuits and systems based upon the operational and organizational principles of biological systems at Caltec in late 1980s [1]. The term “neuromorphic” coined by Carver Mead refers to analog VLSI Artificial neural systems based on biological nervous systems. The Neuromorphic Circuits provide much higher efficiency in computation of large data in parallel as compared to the digitally simulated Neuronal networks and the traditional digital computation systems based on Von Neumann architecture. These systems are suitable for real time large scale neural emulations which is not possible in the other computing systems. Further, the neuromorphic circuits in a VLSI are much compact and consume less power as compared to their digital counterparts [2]. The large scale dense, compact, and low power neuromorphic circuits find their applications in implants and prosthetics.

A group of researchers in the field of Neuromorphic Engineering are working on the implementation of compact low power circuits operating in real time which can be integrated with the biological system. The pioneering work in this field has been done by Giacomo Indiveri, Institute of Neuroinformatics at Zurich. The basic components building up these systems are the neurons and synapses analogous to the biological systems. The use of subthreshold regime of the devices in these circuits results into very low power consumption in the circuits. The innovative field of Neuromorphic Engineering thus utilizes the combination of power and area efficiencies of analog circuits with programmability of digital architectures to build up robust and implantable systems. The implementation of such large-scale hardware neuromorphic systems need dense packaging of neurons and synapses to achieve volumes comparable to biological networks in human brain. The challenge therefore is in developing compact and low power neuron and synapse circuits which can mimic the functionality of their biological counterparts.

The neurons are the basic signaling units of the nervous system and has four main regions to its structure: Cell body or Soma, dendrites, axons, and presynaptic terminals. The cell body of the neuron integrates

# Link-Quality Aware Path Selection in the Presence of Proactive Jamming in Fallible Wireless Sensor Networks

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and Sudip Misra, *Senior Member, IEEE*

**Abstract**—In this paper, we propose a mechanism to ensure the proper functioning of a wireless sensor network (WSN) in the presence of static and proactive jammer within the network. Existing research works have primarily focused on the detection of jammer node within the network and ameliorating its consequent effects on the network. However, these countermeasures to mitigate the effect of the jammer suffer from certain limitations as WSNs are primarily resource constrained and most of the countermeasures are computationally intensive. The objective of this paper is to prevent the disruption of the network in the presence of jamming by bypassing the jammed zone and setting up alternative paths. The alternative paths are chosen with the maximum link quality in order to maintain the quality of service of the network even after jamming. This paper proposes *Link-quality Aware Path Selection (LAPSE)* algorithm that chooses alternative paths based on the optimal link quality. LAPSE is based on *optimal decision rule* and its design considers the fallible nature of the nodes while choosing/rejecting a particular link. Finally, the performance of the proposed algorithm, LAPSE, is evaluated in terms of the network parameters—packet delivery rate, network throughput, transmission energy, node lifetime, and network lifetime. Results indicate that the performance of LAPSE is significantly better than the existing jamming avoidance algorithms.

**Index Terms**—Jamming, wireless sensor network (WSN), routing, link quality, fallibility.

## I. INTRODUCTION

IN THE recent past, Wireless Sensor Networks (WSNs) have rapidly emerged and evolved because of their widespread and varied applicability across different domains [1], [2] such as target tracking [3], [4], military applications [5], [6], fuel management [7], health monitoring [8], robotics [9], and agriculture [10], [11]. However, WSNs are also vulnerable to several threats and attacks. Among these, *jamming* is a serious form of attack that disrupts and collapses a WSN very deeply. A jammer

successfully prevents communication within the network by emitting radio signals in the same frequency as that of the nodes operating in the network. This, in turn, causes tremendous increase in the rate of packet drops and also results in very high number of unsuccessful packet re-transmissions. Also, as WSNs are heavily resource-constrained in nature, jamming in WSNs results in faster depletion of the residual energy level of the nodes and an eventual network failure.

In this work, we investigate the problem of avoiding the *jammer-affected* zone of a WSN and address the problem of re-establishing routing connectivity among the other non-jammed nodes of the network. We present the underlying motivation and contribution of this work in the following subsections.

### A. Motivation

Recent research has already proposed several techniques to detect jammers [12], [13] and ameliorate the consequent effects [14]–[16]. However, in each of the countermeasure techniques, there are some difficulties while practically implementing them in a real-life situation involving a static WSN, in the presence of a proactive-jammer. As per the work of Mpitzopoulos *et al.* [17], the difficulties encountered with the proposed countermeasures are briefly discussed as follows:

- *Regulated Transmitted Power:* This technique requires a sensor node to increase the power of the transmitter as a resistant measure against the jammer. However, such nodes with the ability to regulate their transmission power is highly rare and vendor-specific, thereby questioning the generality of applicability of this solution.
- *Spread Spectrum Methods:* In Frequency-Hopping Spread Spectrum (FHSS) transmission, the sensor node must possess compatible hardware to enable frequency switching and a very high bandwidth is required, compared to single frequency transmission. Another approach is the usage of the Direct Sequence Spread Spectrum (DSSS) method to detect jamming. As mentioned in the work of Spuhler *et al.* [12], the work focuses on the estimation of the packet delivery ratio. From the estimated value of the ratio, the work aims to detect the presence of jammers. The work targets to take the advantage of IEEE 802.15.4 standard by which it is possible to study the preamble for

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## REDUCTION OF MUTUAL COUPLING BETWEEN CLOSELY SPACED PATCH ANTENNAS USING METAMATERIAL

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□

**Abstract**—Constraining the surface waves which are present in patch antenna array using Metamaterials is one of the ways to reduce mutual coupling in patch antenna array. This concept is shown by the design of Edge Located Via Slot Type Electromagnetic Band-Gap (EVS-EBG) structure in this paper. In the first part of the paper identification of EVS-EBG is done using reflection phase diagram. In the second part a compact patch antenna array is designed and application of EVS-EBG for reduction of mutual coupling is shown. Substrate which is used for designing EVS-EBG structure is FR-4 epoxy whose dielectric constant is 4.4 and thickness of the substrate is 1.6 mm. Reduction of mutual coupling between patch antenna array using EVS-EBG structure is achieved at 5.8 GHz-frequency which is a ISM band. Simulation of the EVS-EBG structure is done using Ansoft HFSS. The simulated results are verified by measuring fabricated prototype. Simulation results are showing 11 dB mutual coupling reduction and laboratory results are showing 10.11dB mutual coupling reduction when two lines of EVS-EBG is inserted between patch antennas.

**Index Terms**—Bandgap, Reflection Phase Diagram, Surface waves.

### I. INTRODUCTION

Researcher in Physics and Antenna community has shown great amount of interest in studying Metamaterial and its applications in past few years. Metamaterials find applications in sound filtering, Seismic protection, Cloaking devices, Absorber and also for reducing RCS in Radar. Metamaterial (artificially engineered material) are build using periodic inclusions of metals and dielectric [1]. Metamaterial has ability to influence electromagnetic and optical properties [2]. When metamaterial interact with electromagnetic waves various features like frequency stopband, passband and bandgap could be recognized. EBG structures show unique bandgap properties and therefore are considered as a special type of metamaterial. EBG are 3-D structure or 2-D planar surfaces and are defined as periodic and non-periodic objects that stops the propagation of electromagnetic wave in specified band of frequencies [3]. In optics, photonic bandgap concept is the origin of terminology of electromagnetic bandgap structures which are also made up of periodic structures. Various multilayer electromagnetic bandgap structures which are of diverse form and new bandgap are used in many applications, lot of applications in which undesirable modes and harmonics are controlled are explained [4]. EBG are used for reducing mutual coupling between patch antenna array as it can suppress the flow of surface waves [5]. In [6] the author has designed slot loaded EBG between two planar monopole antennas, the bandgap for the designed structure is centered at 5.5GHz. A compact spiral type EBG was designed by N.Yuan with bandgap centered at

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## Efficient Frequency Reuse Method for LTE Femtocell Network

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### ABSTRACT

Frequency reuse is the basic concept of commercial wireless communications. Frequency reuse is performed by partitioning of radio frequency (RF) area into cells. Each cell is designed to use radio frequencies within its boundaries. Hence the same frequencies can be reused in other cells which are not far away from each other without interference. These cells are known as co-channel cells. In the Long Term Evolution (LTE) integrated with femtocell (FC), deployment density faces the problem of intercell interference and resource allocation during handover. It results in the degradation of throughput of users, increase in delay and the reduction of Quality of Service (QoS). To overcome these issues we proposed a novel method as Efficient Frequency Reuse (EFR) procedure. Efficient Frequency Reuse (EFR) is based on graph-coloring concept. We proposed algorithms as EFR( ) and NEWFRE() for frequency allocation. In our method we considered the dynamic approach instead of fixed one for reusing the frequencies. The proposed scheme enhances the conventional Dynamic Fractional Frequency Reuse (D-FFR) by efficiently allocating the spectrum to cell and cell users. Such allocation can significantly handle traffic load in different cells in practical environment. Hence our proposed EFR method has shown the remarkable improvement for interference mitigation during handover.

### KEYWORDS

Long Term Evolution; 3rd Generation Partnership Project(3GPP); Dynamic Fractional Frequency Reuse(D-FFR); Handover; Interference; Efficient Frequency Reuse (EFR); macrocell; femtocell.

### INTRODUCTION

The increasing demand for higher data rate and coverage area have motivated the femtocell utilization. In the LTE integrated with femtocells (FC) network, high deployment of femtocells faces the problem of intercell interference and resource allocation during handover. The Intercell Interference Coordination (ICIC) is a strategy to control the downlink Intercell Interference (ICI) in OFDM networks. ICIC manages the coordination among the neighboring cells to allocate orthogonal resources to the highly overlapping interfered areas [1].

While in the LTE network integrated with femtocells (FC), deployment density faces the problem of intercell interference and resource allocation during handover. It results in poor performance in terms of throughput of users, delay and the Quality of Service (QoS).

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## Offline signature verification using shape correspondence

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**Abstract:** Biometrics has always been an integral part of human identification and verification, with offline signature verification being a most crucial component of it. It is a challenging task as the signatures are time variant. To address the above difficulty, this paper presents a novel approach to identify the correspondence between pixels of different signatures using an adaptive weighted combination of shape context distance and Euclidean distance. These correspondences are then used for the transformation of query signature plane to reference signature plane using thin plate spline transformation. The distances between signatures are computed using plane transformation, a shape descriptor, and the farness between matched pixels. The computed distances are then fed to the support vector machine (SVM) classifier to determine the merit of genuineness. With the proposed methodology, better accuracy is obtained. The results exhibit an accuracy of 89.58% using proposed method on GPDS synthetic signature database.

**Keywords:** handwritten signature verification; pattern recognition; pattern analysis; shape matching; thin plate spline transformation; shape context; document analysis.

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# Offline Handwritten Signature Verification Using Cylindrical Shape Context

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**Abstract** Offline handwritten signatures is a convincing evidence form of biometrics for verification. However, the verification of offline handwritten signatures is challenging task because of the variations in handwritten signatures. To address this difficulty, this paper proposes a new approach to represent the shape. In this newly proposed approach, the signature pixels are represented by: (1) Gaussian Weighting Based Tangent Angle, to represent the curve angle at the reference pixel; (2) a new shape descriptor, i.e. cylindrical shape context is proposed for a detailed and accurate description of the curve at corresponding pixel. Experimental results show that desired pixel matching results are obtained by using cylindrical shape context which automatically increases the accuracy of verification of offline handwritten signatures. The shape dissimilarity measures are computed and given to the Support Vector Machine with Radial Basis Function (RBF) kernel for classification of signature. The results obtained using GPDS synthetic

signature database, UTSig persian offline signature database, and MCYT-75 offline signature database shows the effectiveness of proposed cylindrical shape context.

**Keywords** Offline signature verification · Cylindrical shape context · Shape descriptor · Biometrics · Shape matching · Shape dissimilarity measures

## 1 Introduction

Signature is the oldest form of biometric for authentication of an individual or a document. Even in the modern era, it is one of the most widely used biometric for the authentication of various important documents such as certificates, contracts, bonds and bank checks. Researchers are trying to automate the process of verification of this huge quantity of documents using handwritten signature.

A handwritten signature is generated by a complex process based on the situations of the signer and psychophysical state under which signature is made, which results dynamic variations in signature. There are natural variations present in the signature made at different timings, even if the signature originates from a well-practiced expert. The intrapersonal similarity, i.e. similarity between the signatures made by the same person is more as compared to the interpersonal

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# Analysis of subthreshold SOI FinFET based two stage OTA for low power

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## Abstract

In this paper, subthreshold design and analysis of Silicon on Insulator Fin Field Effect Transistor (SOI FinFET) based two stage Operational Transconductance Amplifier (OTA) is presented for low power and low supply voltage in nanometre regime. The OTA design optimization is achieved by  $g_m/I_D$  methodology which helps to determine the device aspect ratios. Compactness is achieved by using nanometre FinFET technology. The OTA design is simulated using 30 nm SOI FinFET Berkeley Short-channel IGFET Common Multi-gate (BSIM-CMG) model, with bias current and supply voltage of 20 nA and  $\pm 0.5V$  respectively. The simulation results in subthreshold regime of FinFET based two stage OTA has a gain of 57 dB with a phase margin of 69.81 degree, Common Mode Rejection Ratio (CMRR) of 61.55 dB and power consumption of 108 nW.

**Keywords** SOI FinFET · Subthreshold · Low supply voltage · OTA · Gain · CMRR · Low Power dissipation

## 1 Introduction

The wide spread propagation of passive sensors facilitated by the Internet of things revolution and requirement of low-power biological sensors has spurred the interest in the field of subthreshold region operating devices. These devices must support long battery life as they will be deployed in sensors which may be located in sparsely populated, inaccessible regions of the world. An improvement for new technologies on numerous fronts, both on device and circuit level, systems and applications is necessary. Although continuous down scaling of supply voltage and transistor channel-length has unbelievably run-down the performance of the analog part of the circuit. It can be concluded that the nanometre FinFET device

technology for miniaturization is the best option, for the low power regime in analog domain [1–3]. The analog circuits operating in subthreshold region are used for low voltage and low power application because they have high  $g_m/I_D$  ratio, providing high energy efficiency. Moreover, most of the research work reported in analog domain is based on nanometre CMOS technologies [4–10]. This paper deals with current research in analog domain, related to design of subthreshold FinFET based OTA particularly at low technology node of 30 nm.

The FinFET device is biased in the subthreshold region of operation also known as weak inversion region. This ensures that the FinFET draws very low current and also dissipates negligible amount of power. The design of an ultra-low power highly sensitive even at low voltages analog front end is a must in biomedical applications. Hence our focus was on OTA which is an integral part of any analog circuit. It can be used as a part of the front end of biomedical devices as well as used in passive sensors in remote places. In this paper we investigate and simulate the subthreshold region performance of 30 nm SOI FinFET based two stage Operational Transconductance Amplifier (OTA) with a high- $\kappa$  gate dielectric hafnium oxide ( $HfO_2$ ) for low power and low voltage supply applications. The OTA circuit consists of input differential stage followed by

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## Features

### Design and Analysis of SOI FinFET Based Three Stage OTA with Nested Gm-C Frequency Compensation

<sup>1</sup>99B5\*CB?IG5F9=@5B?5F<sup>6</sup>CG5J= <sup>5</sup>\*IH5F<sup>5</sup>\*\*5H<C8<sup>5</sup>

<sup>5</sup>9D5FHA9BHC:@97HFCB=7G5B8+9@97CAAIB=75H=CB\*5F85F'5H9@ BGH=H1H9C:+97<BC@C;MSIA65= B8=5

<sup>6</sup> B8=5B BGH=H1H9C:+97<BC@C;MCA65MSIA65= B8=5

**Abstract** R BH<=GD5D9F89G=;B5B85B5@MG=GC:\*=@=7CBCB BGI@5HCF=B=9@8::97H+F5BG=GHCF\*& =B+65G98H GH5;9 &D9F5H=CB5@ +F5BG7CB8I7H5B79 AD@=:9F &+ =G DF9G9BH98 K=H< %9GH98 +F5BG7CB8I7H5B79 A 5D5 % F9E19B7M CAD9BG5H=CB A9H<C8 =B B5BCA9HF9 F9;=A9 B5@MG=G G<CKG H<5H % A9H<C8 G=:B=:75BH@M =ADFCJ9G H<9 :F9E19B7M 8CA5=B D9F:CFA5B79G C: HCH5@ 7=F7I=H GI7< 5G D<5G9 A5F;=B '\$ GH56=@=HM 5B8 ; DFC8I7H.K=H<GA5@@@7CAD9BG5H=CB75D57=HCFG9F'9@9M\*<CFH7<5BB9@ +CAACBSI@H=:5H9\* \$\$ AC89@<5G699B1G98:CF5H<F99GH5:9 BA\*& =B+65G98&+7=F7I=HG=AI@5H=CB+<989G=:B985AD@=:9F<5G 8;5=BK=H< 89:F99D<5G9A5F;=BIB=HM;5=B65B8K=8H<C: \$N 3.DCK9F8=GG=D5H=CB:CFP -GIDD@M JC@H5;95B8 D@C5875D57=HCFG9GD97H=J9@M

#### 1. Introduction

+<9 89J9@CDA9BH=B BH9;F5H98 =F7I=H H97<BC@C;M <5G F9GI@H98 =B G75@=B; 8CKB C: 89J=79 G=N9G : JC@H5;9G CK9J9F H<F9G<C@8 JC@H5;9G 5F9 BCH G75@98 8CKB DFCDCFH=CB5@@M 5H H<9 G5A9 F5H9 +<=G 7@ =B7F95G=B;89A5B8:CF@CKDCK9DFCFH56@965HH9FMCD9F5H989@97HFCB=789J=79G@=79AC6=@9D<CB9G5B8@ H<9 GH=AI@IG :CF :IFH<9F F9G95F7< HCK5F8G 57<=9J=B; <=:<9F CB 7<=D =BH9;F5H=CB 5B8 @CK DCK9F 7CBGIA F9DCFH98=BH<9@=H9F5HIF9H<5HH<9B5BCA9HF9=B+89J=79H97<BC@C;M:CFA=B=5HIF=N5H=CB=GH<969GHCDI DCK9FF9;=A9=B5B5@C;8CA5=B2 3 BDCFH56@989J=79G<=:<5=BK=8965B8K=8H<5AD@=:9FG8F=J=B;@5F;975D57: @C58G G9FJ9'5G 9FFCF 5AD@=:9FG =B @CKJC@H5;9 @=B95F F9;I@5HCFG 9B79 H<9F9 =G 5 89A5B8 :CF @CK DC :F9E19B7M7CAD9BG5H98@H=GH5;AD@=:9KG=7<5F9 75D56@C: 8F=J=B;@5F;975D57=H=J@C58CG=:B 5B8 CDH=A=N5H=CB C: AI@H=GH5;9 CD9F5H=CB5@ 5AD@=:9FG =G 697CA=B;=B7F95G=B;@M 7<5@@9B;956@9 =B : SCF9CJ9FACGHC:H<9F9G95F7<KCF?F9DCFH98=B5B5@C;8CA5=B=G65G98CBB5BCA9HF9S&\*H97<BC@C;=9G2 D5D9F895@GK=H<7IFF9BHF9G95F7<=B5B5@C;8CA5=BF9@5H98HC89G=:BC:\*& =B+65G98H<F99GH5;9&+K=H<B9C A:F9E19B7M7CAD9BG5H=CB5H BAH97<BC@C;MBC89

BCF89FHC89G=:B&+7=F7I=HK=H<=:<5=B5B8K=8965B8K=8H<5HH<9G5A9H=A9AI@H=GH5;9=GB979GG5FM+ =B7F95G9=B588=H=CB5@GH5;9G75B699=H<9F75G75898CF75G7C898+<975G7C898GHFI7HIF9F9EI=F9GAI@H=D@9J C: HF5BG=GHCF 69HK99B H<9 GIDD@M JC@H5;9G K<=7< =B7F95G9G H<9 <958FCCA C: H<9 7=F7I=H +<=G <5G DF :C7IG CB =AD@9A9BH5H=CB C: 75G75898 GHFI7HIF9G :CF <=:<5=B SI@H=GH5;9 &+G 5F9 65G98 CB 75G758=B; C: <CF=NCBH5@@@MHC=ADFCJ9H<9;5=B5B8JC@H5;9GK=B;6IH=H=G57<=9J985H5B9LD9BG9C:65B8K=8H<K<=9CD9 GIDD@MJC@H5;9+<9;5=B65B8K=8H<C:5B5AD@=:9F=GJ9FMAI7<89D9B89BHCBFH5BG7CB8I7H5B795B875D57=H5 57H=J9 89J=79 6CH< C: K<=7< 5F9 =BHF=BG=7 D5F5A9H9FG C: H<9 89J=79 +<9 AI@H=D@9 DC@9 B5HIF9 C: AI@H IB89F;C9G7@CG98@CCDGH56=@=HMDFC6@9AG9B797CAD9BG5H=CB=GIG98HC=ADFCJ9:F9E19B7M7MGH56=@=HM A5?9 :5=B 5B8 D<5G9 @=B95F :CF F9EI=F98 :F9E19B7M F5B;9 S5BM DCGG=6@9 5@H9FB5H=J9G C: 7CAD9BG5H=CB F9DCFH98 =B H<9 @=H9F5HIF9 HC =ADFCJ9 GH56=@=HM :F9E19B7M 5B8HF5BG=9BH F9GDCBG9G C: H<9 AI@H= :CF9ACGH 6CHH@9B97? C: GI7< 5AD@=:9FG =G H<5H H<9 DC@9 5H H<9 CIHDIH =G DI@@@98 =BG=89 J9FM 7@C K<=7<G=:B=:75BH@M@=A=HGH<965B8K=8H<5B8A5?9GH<97CAD9BG5H=CB C: H<95AD@=:9F8=:7I@HIB89F@ 5F95 7CBGHF5=BH CAD9BG5H=CB IG=B; 75D57=HCFG CF IG=B; :998 :CFK5F8 D5H<G 5F9 H<9 ACGH 7CAACB@M I 7CAD9BG5H=CB H97<B=EI9G =B AI@H=GH5;9 5AD@=:9FG 2 3 6CJ9 5@@ 5 :CC8 HF589C:: 69HK99B ;5=B 5B8 7CAI :J9B6M75G758=B;H<F99G=AD@9GH5;9G BH<=GD5D9F9K9DFCJ=895K9@@89=:B9889G=:BDFC798IF9G=BK<=7<H<9 F5H=CC:957<HF5BG=GHCF5B8H<9=F6=5G7IFF9BHG5F95B5@MH=75@@MF9@5H98HCH<9H<F99GH5;9&+GA5=BD: +<9D5D9F=GCF;5B=N985G:C@@@CKG\*97H=CB895@GK=H< BA\*& =B+89J=79GHFI7HIF9=BH9;F5H98=BH<F

&+ 7=F7I=H \*97H=CB :J9G H<9 =BG=:<H=BHC % A9H<C8 IG98 :CF :F9E19B7M 7CAD9BG5H=CB =B H<F99 GH5;9 & 89G=:B 5B8 5B5@MG=G C: =B+ 65G98 H<F99 GH5;9 &+ 7=F7I=H K=H< % 5F9 9LD@5=B98 =B \*97H=CB +<9 7=F7I=H G=AI@5H=CBF9GI@HG5B88=G7IGG=CB C:\*& =B+65G98&+K=H<%5F9DF9G9BH985B857CAD5F=GCBC:H<9DFCDCG 5AD@=:9FK=H<DI6@=G<987CAD9BG5H=CBH97<B=EI9G=:J9B=B\*97H=CB=B5@@M\*97H=CB7CB7@189GH<9KCF

# Trust based mechanism for Securing IoT Routing Protocol RPL against Wormhole & Grayhole Attacks

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**Abstract**— Internet of Things is attracting a lot of interest in the modern world and has become a part of daily life leading to a large scale of distribution of Low power and Lossy Networks (LLN). For such networks constrained by low power and storage, IETF has proposed RPL an open standard routing protocol. However RPL protocol is exposed to a number of attacks which may degrade the performance and resources of the network leading to incorrect output. In this paper, to address Wormhole and Grayhole attack we propose a light weight Trust based mechanism. The proposed method uses direct trust which is computed based on node properties and Indirect Trust which is based on opinion of the neighboring nodes. The proposed method is energy friendly and does not impose excessive overhead on network traffic.

**Keywords**—IoT, RPL, Trust, Wormhole, Grayhole, Direct Trust, Indirect Trust.

## I. INTRODUCTION

Internet of Things (IoT) is a dynamic global ecosystem where billions of small devices with data capture and communication capabilities are seamlessly integrated into the information network [1]. IoT is creating a new epoch of innovation that connects the digital and machine ecosystems and brings speed and effectiveness to various industrial sectors and the daily human life. These devices have self-configuring capabilities based on interoperable communication protocols. Using intelligent interfaces these devices interact with "smart things" over the internet. Experts forecast by the year 2020, 20-50 billion IoT devices will be part of the internet in the areas of healthcare, automotive, security, manufacturing, services, energy, agriculture, smart homes and retail utilities according to Gartner report [2].

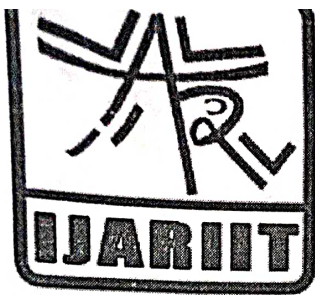
The fact that these devices can exchange information with one another and over the web poses a security risk to the sensitive information increasingly being available online and more endpoints exposed to attackers. Hence IoT security has become a main concern in the research community and industrial world. Gartner 2016 [2] article identifies IoT security as the top ten technologies for 2017-18. However these devices have limited data storage and processing capabilities, to add the devices have limited power stored in tiny batteries. Hence security challenges with regards to data,

privacy, confidentiality, identification and access control are pertinent.

In the IoT revolution, the security challenges needs to be addressed in order to build confidence in public and achieve global success with IoT. Routing of data packets among IoT devices is an exposed area. International Engineering Task Force (IETF) has standardized the Routing Protocol for Low Power and Lossy Network (RPL) as routing protocol for IoT networks [4]. Like any other wireless routing protocol, this protocol is also exposed to numerous attacks [5]. The attacks can be categorised as attacks related to network traffic, attacks consuming network resources and attacks altering the network topology. IoT networks topology and network paths can be destabilized by these attacks. The authors of [6] investigated the protection capabilities of RPL protocol against these attacks and found that RPL doesn't have self healing mechanism to prevent these attacks except HELLO flood attack, hence some security mechanism is needed.

The proposed method seeks to address suboptimisation attack called Wormhole and isolation attack called Grayhole attack as these attacks can endanger the stability of IoT networks. As the IoT devices are provided with fewer resources the security mechanism should be less resource hungry and simple so that it will not require more computational power or memory power. Hence a Lightweight Trust mechanism is proposed for securing RPL against Worm Hole and Grayhole attacks. The proposed mechanism is energy affable and does not compel undue overhead on network traffic.

The remaining paper is structured as follows: Section II gives a brief introduction of RPL protocol. Section III discusses wormhole and grayhole attacks and describes the countermeasures to contain the attack. Section IV introduces the proposed Trust based mechanism. Section V presents simulation results using Cooja/ Contiki environment and demonstrates the effectiveness of proposed Trust based mechanism in terms improved network performances. Section VI concludes the work done and final notes on the future scope.



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## Detecting man in the middle attack

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### ABSTRACT

With Cyber-attacks and cyber-threats are increasing, network security needs to be seen in a new dimension. WLANs (Wireless Local Area Networks) are acquiring their hold in all the verticals of life. As technology is growing, a data breach at any scale, whether at the industry level or private can be menacing to distinct. Man-in-the-Middle is one of the most ruinous attacks in the WLAN. In this work, I propose a solution to detect a Man-in-the-Middle attack in public or home WLANs. The objective of the work is to present a solution which allows individual to keep watch on WLAN based on data packets collected over the network and evaluating its behavior. The proposed detection system is simulated with the help of Kali Linux where the attacker is trying to get a WPA Handshake over the network, a packet sniffer tool which captures the changes over the network during the simulation and one capture analyzing tool which analyze the capture.

**Keywords:** Kali Linux, MITM, Network Scanners, Brute force

### 1. INTRODUCTION

Wi-Fi connections are all over the place now, whether you are at your home or your favorite coffee shop. Users are using Wi-Fi for paying their bills, buying tickets or transferring money from one to another. In these scenarios, the threat of data theft can prove a loss of money, status or dependence. Cyber-attacks are on the upswing and cyber-crime is here to stay in today's digital era. Security is a major concern for WLANs because of its broadcast nature in communication. Every data packet sent in a wireless medium is broadcast by nature where any user in the communication range can capture the packet. We might feel that by installing a firewall and sitting on the safe side of the network guarding you against the world of the internet is the best defense against cyber-criminals. Nevertheless, the greatest risk to computers and networks today comes not from the front door through the firewall. The only measure most people use to secure their home network is to set up a password and prevent other people from taking control over your data. A grim risk is that an online immoral might exploit your poor Wi-Fi security measures. Through this, he can "listen" to your traffic in order to retrieve sensitive material or take advantage of your network to launch

malicious attacks such as Man-in-The-Middle attacks, network sniffing or data theft. Approximately 25% of Wi-Fi hotspots in the world do not use any encryption at all. The effort required to hack a WLAN is less when network setting are set unsuitably and predictable passwords are set by the owner.

Data loss in communication is resulted due to many factors some of them includes data being corrupted or made unreadable by the software or application. Data loss is also referred to as Data leakage. Data loss causes the network to load resources at a slower pace and most of the time it is due to the bad setup of network or wired connection to the router. But when the scenario is not that, it is susceptible for a user to recheck the network behavior and do an analysis to surf safely over the internet. One of the many scenarios and the daunting is network getting attack by Man-in-The-Middle attack. In MITM the network gets compromised by the hacker which is capturing the packets passing over the channel. Those packets can be decrypted by passing them through brute-force or social engineering. A man-in-the-middle attack can be done in many ways, one of the basic and most effectual ways is to get a WPA Handshake. In WPA Handshake the attacker gets a three-way handshake by deauthorizing the station and the client. Here station refers to the router and client refers to the authorized user connected to the router.



Fig. 1: Secured connection over the network

In the above fig. the Client - A is connected over the network with Host -B making a secured connection over the network. The transaction is only between these two nodes and

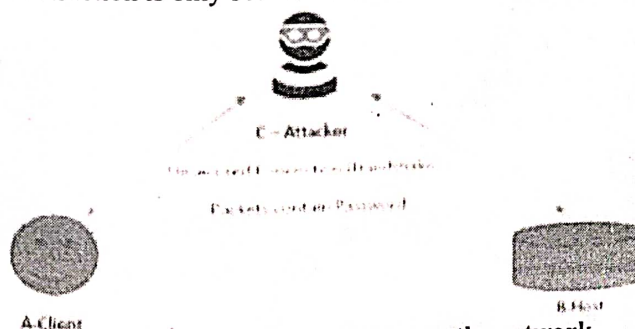


Fig. 2: Unsecured Connection over the network

# Weight Assignment Algorithms for Designing Fully Connected Neural Network

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**Abstract**—Soft computing is used to solve the problems where input data is incomplete or imprecise. This paper demonstrate designing fully connected neural network system using four different weight calculation algorithms. Input data for weight calculation is constructed in the matrix format based on the pairwise comparison of input constraints. This comparison is performed using saaty's method. This input matrix helps to build judgment between several individuals, forming a single judgment. Algorithm considered here are Geometric average mean, Linear algebra calculation, Successive matrix squaring method, and analytical hierarchical processing method. Based on the quality parameter of performance, it is observed that analytical hierarchical processing is the most promising mathematical method for finding appropriate weight. Analytical hierarchical processing works on structuration of the problem into sub problems, Hence it the most prominent method for weight calculation in fully connected NN.

**Index Terms**—Soft Computing, Neural Network, Saaty's Method, Analytical Hierarchical Processing, Exact Linear Algebra Calculation, Geometric Average Approximation, Successive Matrix Squaring.

## I. INTRODUCTION

Finding a solution to a real time problem is difficult as it may depend on the type of problem, sensitivity of the problem, and type of solution expected. Soft computing approach helps to find a solution in unpredictable situation. Neural network (NN) is one of the techniques of the soft techniques. NN requires weights to be assigned among neurons for calculating result. This paper discusses 4 different algorithms for calculating weights based on input given to the neurons. This paper also compares stated algorithm with an example.

### A. Soft Computing Approach

Using computer technology, a process which completes a task is called as computing. Computing is classified into two types. First is Hard computing, i.e.,

conventional computing concept which uses the available analytical calculation model. Second is Soft computing, which works as a tolerant of imprecision, uncertainty, partial truth, and approximation. [28] It describes and transforms information such as theory, analysis, design, efficiency, implementation, and application as that of the human mind. [1] [25] The main characteristic of soft computing is its intrinsic capability to create hybrid systems based on a (loose or tight) integration of the constituent technologies. [5] This integration combines unclear domain knowledge and empirical data to develop flexible computing tools and to solve complex problems based on the reasoning. [25][8][16] Soft computing is more oriented towards the analysis and design of intelligent systems.

Soft computing (SC) has a different technique like neural networks, fuzzy logic, genetic algorithm, swarm optimisation and their hybrid combination. [9] SC has strong learning and cognitive ability and good tolerance of uncertainty and imprecision. SC techniques derive their power of generalisation generating output from previously unseen or learned inputs based on approximation. [25] Soft Computing algorithms are used for real time understandable application.

Advantages of Soft Computing can be listed as

- Nonlinear problems can be solved using SC;
- It works in human knowledge areas such as cognition, recognition, understanding, learning, and computing.
- Intelligent systems such as autonomous self-tuning systems, and automated designed systems can be constructed using SC.

## II. STRUCTURE OF THE NEURAL NETWORK

### A. Artificial Neural Network

The artificial neural network is one of the main techniques of soft computing. [20] [23] [5] NN are nonlinear predictive models that learn through experience